

Lexicographic Trust-Region Guidance for Localized Diffusion Counterfactuals

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Abstract

Image counterfactuals should change an explained decision while preserving identity, unrelated attributes, and evidence outside a declared semantic region. A permanent weighted loss cannot express this priority reliably: preservation may block the target, while aggressive target guidance can cause collateral drift. We present lexicographic trust-region causal concept intervention (TR-CCI), a training-free controller for blended latent diffusion. At reliable denoising steps, TR-CCI decodes a predicted-clean latent, requests attainable target progress, minimizes the best achievable identity/locality violation, and only then minimizes drift over all 39 non-target CelebA attributes. A preservation-aware latent restoration stage uses the same ordering and accepts deterministic backtracking steps without additional U-Net or scheduler transitions. In a completed paired study of 200 unique smile-removal sources, adaptive A11 reduces non-target drift by 11.8% relative to matched fixed A10 (0.0635 versus 0.0720) at nearly matched target-pass rate at the declared 0.8 requirement (37.5% versus 38.5%). Identity and outside-mask error remain close, with additional runtime. Two planned 300-image end-to-end attacked evaluations are specified but deliberately left pending. These results support lexicographic adaptation as a practical way to reduce collateral classifier movement, not as evidence of real-world causal identification.

Keywords: visual counterfactuals, diffusion guidance, trust-region optimization, classifier explanations, localized editing

1 Introduction

A visual counterfactual explains a model decision by showing an alternative image that produces a desired outcome. For faces, changing the target alone is not sufficient: an explanation should preserve the person, avoid unrelated attribute changes, and confine the intervention to a declared facial region. Otherwise, a classifier can cross its boundary for the wrong reason.

Diffusion models provide a strong image prior for counterfactual generation. DiME and ACE combine classifier objectives with diffusion-based image regularization [8, 9]; DiG-IN optimizes diffusion variables for model-analysis objectives [1]; and MaskDiME adapts localization during generation [6]. Blended latent diffusion (BLD) further

preserves the source outside a spatial mask by restoring a correspondingly noised source latent after each scheduler step [2]. These ingredients make localized counterfactuals possible, but do not determine how competing objectives should be coordinated.

The usual solution is a weighted sum,

$$L = w_T L_T + w_D L_D + w_I L_I + w_L L_L, \quad (1)$$

where target, non-target drift, identity, and locality use different models, scales, and denoising-time geometries. One fixed set of weights must then serve two incompatible states: first reach the target; later preserve or restore the remaining evidence. Large preservation weights can prevent a counterfactual, whereas weak weights admit collateral drift. The effect of a latent step also changes with scheduler time, making noise-space clipping difficult to interpret.

We instead impose a lexicographic local order. Target progress is primary; identity and outside-mask locality define the best attainable safety envelope; and non-target attribute drift is optimized only inside that envelope. Each guided step solves a small trust-region problem in predicted-clean latent coordinates. Its radius adapts from achieved rather than merely requested target progress. A final restoration stage applies the same rule to the actual post-BLD latent.

This paper makes three contributions:

- predicted-clean lexicographic trust-region guidance integrated into BLD without differentiating through the U-Net;
- a matched fixed/adaptive comparison in which A10 and A11 share the mask, scheduler, clean-coordinate trust budget, and final-optimization budget;
- preservation-aware final latent restoration with deterministic backtracking, a cumulative radius, and an auditable acceptance trace.

The term “causal concept intervention” denotes a controlled intervention on the evidence used by the explained classifier. It does not assert an identified causal effect in the physical world.

2 Related Work

Diffusion counterfactuals. Generative counterfactual methods use structured latent spaces to produce

plausible alternatives [14, 7]. DiME, ACE, TIME, and ECED adapt diffusion generation to visual counterfactual explanations [8, 9, 10, 13]. DiG-IN optimizes diffusion inputs end to end, while a prior facial-counterfactual pipeline combines face-part masks, diffusion generation, and adversarial analysis [1, 18]. Our focus is narrower: coordinating a target, two safety constraints, and continuous non-target preservation inside a fixed localized diffusion pipeline.

Predicted-clean guidance and localization. Latent diffusion performs sampling in an autoencoder space [15], and DDIM defines the scheduler used here [17]. Universal Guidance and diffusion posterior sampling show that external measurements can act on predicted-clean estimates rather than on noisy states [3, 5]. MaskDiME uses classifier information to adapt an edit mask [6]; in contrast, we freeze semantic support before generation and adapt the objective geometry inside that support.

Multiple gradient objectives. Gradient-surgery methods alter conflicting multi-task directions [19]. A weighted or projected direction still leaves the priority policy implicit. TR-CCI expresses it directly as sequential local subproblems: attain target progress, minimize unavoidable safety violation, then reduce collateral drift.

3 Lexicographic Trust-Region CCI

3.1 Localized Counterfactual Objective

Let $x_s \in [0, 1]^{3 \times H \times W}$ be a source image, $f_j(x)$ the logit of attribute j , T the target index, and $y_T \in \{0, 1\}$ the desired target value. For required desired probability p^* , define

$$\kappa = \log \frac{p^*}{1 - p^*}, \quad c_T(z) = \kappa - (2y_T - 1)f_T(x(z)). \quad (2)$$

The target is feasible when $c_T(z) \leq 0$.

Two aligned masks have different roles. A binary semantic mask m_s states where the intervention is allowed to count; a dilated, feathered generation mask $m_g \in [0, 1]$ supplies smooth latent support and source blending. The primary preservation objective includes every non-target classifier output:

$$D_{NT}(z) = \frac{1}{K-1} \sum_{j \neq T} \text{Huber}_\delta(p_j(x(z)) - p_j(x_s)). \quad (3)$$

No correlated attribute is exempted. Identity and locality define the safety residuals

$$c_I(z) = 1 - \cos(e(x(z)), e(x_s)) - \varepsilon_I, \quad (4)$$

$$c_L(z) = \text{mean}((1 - m_g)|x(z) - x_s|) - \varepsilon_L. \quad (5)$$

Identity uses a frozen FaceNet representation trained on VGGFace2 [16, 4]. Residual total variation is recorded for audit but is not optimized by the trust-region controller.

3.2 Predicted-Clean Measurements

At denoising step t , let z_t be the current latent, $\epsilon_\theta(z_t, t)$ the detached U-Net prediction, and $\bar{\alpha}_t$ the scheduler cumulative alpha. For epsilon prediction,

$$\hat{z}_0 = \frac{z_t - \sqrt{1 - \bar{\alpha}_t} \epsilon_\theta(z_t, t)}{\sqrt{\bar{\alpha}_t}}, \quad (6)$$

$$\hat{x}_0 = D(\hat{z}_0/s_{VAE}), \quad s_{VAE} = 0.18215. \quad (7)$$

The target, drift, identity, and locality models evaluate $\hat{x}_0$. Their image-space measurement gradients pass through the fixed VAE decoder,

$$g = J_D(\hat{z}_0)^\top \nabla_x L, \quad (8)$$

but not through the U-Net. Thus the decoder is a differentiable measurement map; it does not perform another diffusion transition.

3.3 Lexicographic Trust-Region Update

Let $g_T = \nabla c_T$, $g_D = \nabla D_{NT}$, $g_I = \nabla c_I$, and $g_L = \nabla c_L$ with respect to $\hat{z}_0$. Effective generation support is applied before norms, inner products, or optimization. When the target is infeasible, the requested local progress is

$$q_t = \min(\max(c_T, 0), \chi \Delta_t \|g_T\|_2), \quad (9)$$

where Δ_t is the clean-latent trust radius and $0 < \chi < 1$.

The first subproblem finds the smallest unavoidable worst safety residual:

$$\tau^* = \min_{d, \tau} \tau \quad (10)$$

$$\begin{aligned} \text{s.t. } & g_T^\top d \leq -q_t, \\ & c_I + g_I^\top d \leq \tau, \quad c_L + g_L^\top d \leq \tau, \\ & \|d\|_2 \leq \Delta_t. \end{aligned} \quad (11)$$

Inside that best envelope, the second subproblem minimizes collateral drift:

$$d^* = \arg \min_d g_D^\top d + \frac{1}{2\eta_t} \|d\|_2^2 \quad (12)$$

$$\begin{aligned} \text{s.t. } & g_T^\top d \leq -q_t, \\ & c_I + g_I^\top d \leq \tau^* + \zeta, \\ & c_L + g_L^\top d \leq \tau^* + \zeta, \\ & \|d\|_2 \leq \Delta_t. \end{aligned} \quad (13)$$

If the target is already feasible, the progress constraint becomes the guard $c_T + g_T^\top d \leq 0$. If the requested progress is infeasible, deterministic bisection reduces q_t ; an unreliable target gradient skips the update.

Only four masked gradients participate. The implementation therefore solves the subproblems by deterministic active-set enumeration in their gradient span, using a small CPU float64 Gram matrix. KKT multipliers and residuals are recorded at every active step. The radius is

halved when actual target progress is below one quarter of the accepted request and may grow by $1.5\times$ when more than three quarters is realized at the radius boundary.

The requested clean displacement maps exactly to epsilon-prediction space:

$$\Delta\epsilon_t = -\sqrt{\frac{\bar{\alpha}_t}{1-\bar{\alpha}_t}} d^*. \quad (14)$$

After the scheduler step, BLD restores source content outside m_g :

$$z_{t-1} \leftarrow m_g \odot z_{t-1} + (1 - m_g) \odot z_{t-1}^{\text{src}}. \quad (15)$$

The trace records displacement before and after this blend. A10 uses the same measurement path, constraints, mask support, trust radius, and final budget, but starts from a frozen nominal combination of target, safety, and drift gradients. A11 instead uses Eq. (13), so its effective coefficients follow current residuals and gradient geometry.

3.4 Preservation-Aware Final Restoration

The final BLD latent may remain below the target threshold because a local step is attenuated by the scheduler or source blend. A11 therefore runs at most 12 restoration iterations on the actual post-blend latent. Each iteration recomputes Eqs. (11)–(13) through the fixed decoder and tests fractions $\{1, 1/2, 1/4, 1/8\}$ of the proposed step.

When the target is infeasible, an accepted candidate must strictly improve its desired probability and must not worsen the best identity/locality envelope beyond numerical tolerance. Once target and safety are feasible, a candidate must maintain them and strictly decrease non-target drift. All accepted steps must remain within a cumulative clean-latent radius. Failure of every fraction ends restoration. The saved uint8 image is evaluated again because VAE decoding and quantization can move a boundary-adjacent output.

4 Experimental Protocol

4.1 Completed Clean Study

We evaluate smile removal (`Smiling 1` $\rightarrow$ `0`) on CelebAMask-HQ [12, 11]. The semantic support is a fixed perioral union of mouth, upper-lip, and lower-lip components; a dilated soft generation mask is used for gradients and BLD blending. Two random cohorts contain 100 completed sources each. Their 200 source IDs are disjoint across cohorts and paired across methods. Each source produces one output; no candidate reranking or post-generation attack is used.

All arms use the local Stable Diffusion 2.1 base checkpoint, DDIM, 512² resolution, 35 scheduler steps, classifier-free guidance 5.0, blend start fraction 0.25, seed

42, float32 MPS execution, and batch size one. The desired in-loop target probability is 0.8. A11 uses initial/minimum/maximum clean radii 0.15/0.01/0.30, target progress fraction 0.9, reliability alpha threshold 0.10, Huber transition 0.02, at most 12 final iterations, and cumulative final radius 1.8.

We compare:

- **A0 BLD**: source blending with no CCI update;
- **A10 fixed**: a fixed nominal gradient composition with the same clean-coordinate, constraint, and final-restoration budgets;
- **A11 adaptive**: the proposed lexicographic optimizer.

Metrics. Target@0.8 is the fraction reaching the declared desired probability 0.8 under the explained classifier; it is stricter than flip rate at the 0.5 decision boundary. Identity is the source/output FaceNet cosine. Outside MAE is the strict outside-mask RGB error on the $[0, 255]$ scale. NT drift is the mean absolute source/output probability change over all 39 non-target attributes; unlike Eq. (3), it is reported without Huber smoothing. Runtime is the median per-image wall time. FR and NT drift use the same frozen multi-label classifier that guides A10/A11. This is deliberate explained-model evaluation, not independent-oracle validation.

4.2 Planned End-to-End Attacked Study

The pending end-to-end protocol compares A0 and A11 on 300 paired images with mouth-only support, then on the same 300 IDs with mouth, upper-lip, and lower-lip support. It includes the localized smooth boundary attack and will report FID, spatial FID (sFID), face verification accuracy (FVA), feature similarity (FS), mean number of altered concepts (MNAC), collateral distance (CD), counterfactual output score (COUT), and FR. The same frozen multi-label classifier supplies COUT, FR, MNAC, and CD, as well as attack and CCI guidance. Partial-cohort values are not inserted into this manuscript.

5 Results

5.1 Clean A0/A10/A11 Comparison

Table 1 shows the intended fixed/adaptive comparison. A11 lowers NT drift by 11.8% relative to A10 (0.0635 versus 0.0720) while its Target@0.8 rate is one percentage point lower (37.5% versus 38.5%). Mean identity differs by 0.0034 and outside MAE by 0.0057 on the $[0, 255]$ scale. Thus the saved cohort supports reduced collateral movement at a closely matched, but not identical, target operating point. It does not establish statistical superiority because this table does not include a paired confidence interval.

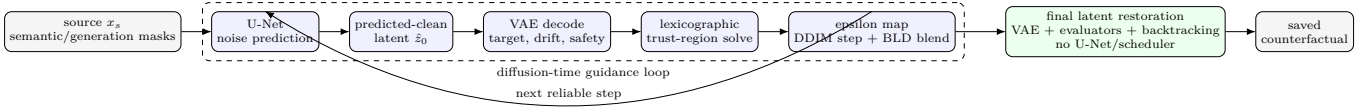


Figure 1: Method overview. A11 evaluates target and preservation on a predicted-clean decode, solves a clean-latent lexicographic trust-region problem, maps the accepted displacement to epsilon prediction, and continues DDIM plus source blending. Final restoration is a distinct post-denoising latent optimization; it reuses the fixed VAE and evaluators but performs no new U-Net or scheduler transition.

Table 1: Completed clean smile-removal comparison over 200 unique paired sources. Target@0.8 is in percent; Outside is strict outside-mask MAE on $[0, 255]$; Time is median seconds per image. Lower NT drift, Outside, and Time are better; higher Target@0.8 and Identity are better.

Method	Target@0.8 $\uparrow$	Identity $\uparrow$	Outside $\downarrow$	NT drift $\downarrow$	Time $\downarrow$
A0 BLD	17.5	0.8757	4.340	0.0477	20.4
A10 fixed CCI	38.5	0.9255	4.334	0.0720	81.8
A11 adaptive CCI	37.5	0.9221	4.339	0.0635	85.9

Table 2: Mechanism ablation for source 26811. This single case illustrates what restoration can do; it is not population evidence.

Metric	Without	With
Desired probability	0.1046	0.7946
Identity cosine	0.8348	0.8704
NT drift	0.0327	0.0600
Runtime (s)	53.5	85.1

Both guided methods substantially increase Target@0.8 over A0 and also increase mean identity cosine in this cohort. This does not mean A0 was optimized for lower identity; A0 simply performs the substantially cheaper BLD path. A11’s median runtime is 85.9 s versus 81.8 s for A10 and 20.4 s for A0. The extra cost comes from repeated VAE decodes, evaluator forwards/backwards, and candidate evaluations during final restoration; the small active-set solve is not the dominant cost.

5.2 Illustrative Final-Restoration Ablation

For source 26811, restoration accepts 8 steps and raises desired probability from 0.1046 to 0.7946 while identity increases from 0.8348 to 0.8704 (Table 2). NT drift also rises from 0.0327 to 0.0600, exposing the cost of recovering a severely missed target. The final image differs from the no-restoration output by pixel MAE 0.000810, maximum normalized change 0.2941, and 4.04% changed pixels. This example validates the acceptance mechanism and its locality, not an average benefit.

5.3 Pending End-to-End Results

6 Limitations

The completed study covers one facial attribute, one dataset, one diffusion backbone, a fixed perioral policy, and deterministic seed-42 generation. Its 200-image statistics are descriptive and do not include identity-cluster bootstrap intervals. The fixed and adaptive operating points are close but not exactly equal in FR; a calibrated matched-success frontier would support a stronger comparison.

The same classifier guides generation and supplies FR and NT drift. This is appropriate when the object of explanation is that classifier, but it can reward classifier-specific features and cannot establish semantic agreement for another model or for people. The pending COUT, FR, MNAC, and CD metrics share this limitation by design. Independent transfer and human evaluation are future validation, not claims of this draft.

The local linearized problems provide an auditable priority rule, not a global convergence guarantee for a changing non-convex diffusion trajectory. Mask quality also bounds the meaning of locality: segmentation errors may omit relevant evidence or admit unrelated pixels. Runtime is several times that of A0 because VAE/evaluator differentiation and restoration line searches are serial on MPS.

Finally, facial images and identity embeddings are biometric data. This method should be used for consented model auditing with appropriate data protection, not for inferring sensitive traits or modifying identity documents. A model counterfactual must be disclosed as generated evidence about a classifier, not as a plausible real-world causal statement.

Table 3: End-to-end attacked evaluation with mouth-only support. The cohort will contain 300 paired images; partial values are intentionally omitted.

Method	FID ↓	sFID ↓	FVA ↑	FS ↑	MNAC ↓	CD ↓	COUT ↑	FR (%) ↑
A0 BLD								Pending (300-image run)
A11 adaptive CCI								Pending (300-image run)

Table 4: End-to-end attacked evaluation with mouth, upper-lip, and lower-lip support on the same 300 IDs. Partial values are intentionally omitted.

Method	FID ↓	sFID ↓	FVA ↑	FS ↑	MNAC ↓	CD ↓	COUT ↑	FR (%) ↑
A0 BLD								Pending (300-image run)
A11 adaptive CCI								Pending (300-image run)

7 Conclusion

TR-CCI replaces a permanent weighted objective with an explicit local ordering: make attainable target progress, minimize unavoidable identity and locality violation, then reduce non-target drift. Predicted-clean measurements place heterogeneous evaluators on an image estimate, clean-coordinate trust regions remove scheduler-dependent step units, and final restoration applies the same priorities to the actual post-BLD latent. On 200 completed paired smile-removal sources, A11 reduces non-target drift by 11.8% relative to matched fixed A10 at a one-point lower Target@0.8 rate, with nearly unchanged identity and outside-mask error but higher runtime. The pending 300-image attacked studies will determine whether that advantage persists in the full end-to-end protocol.

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